Machine Learning Advanced Project

Daraz-Style Daily Orders Prediction

Supervised Learning • Regression • Linear Regression • Streamlit

CLASS JOURNEY

Iris Classification → **Daraz Regression** → **Deep Learning / ANN** → **RNN & Beyond**

Real-world regression project using a synthetic daily-orders dataset

Where This Project Fits in Our Machine Learning Journey

Classification came first. Regression is our second major supervised-learning task.

1

SUPERVISED CLASSIFICATION

Iris: predict a category / class

2

SUPERVISED REGRESSION

Daraz-style: predict a number

3

DEEP LEARNING

ANN: learn with layers of neurons

4

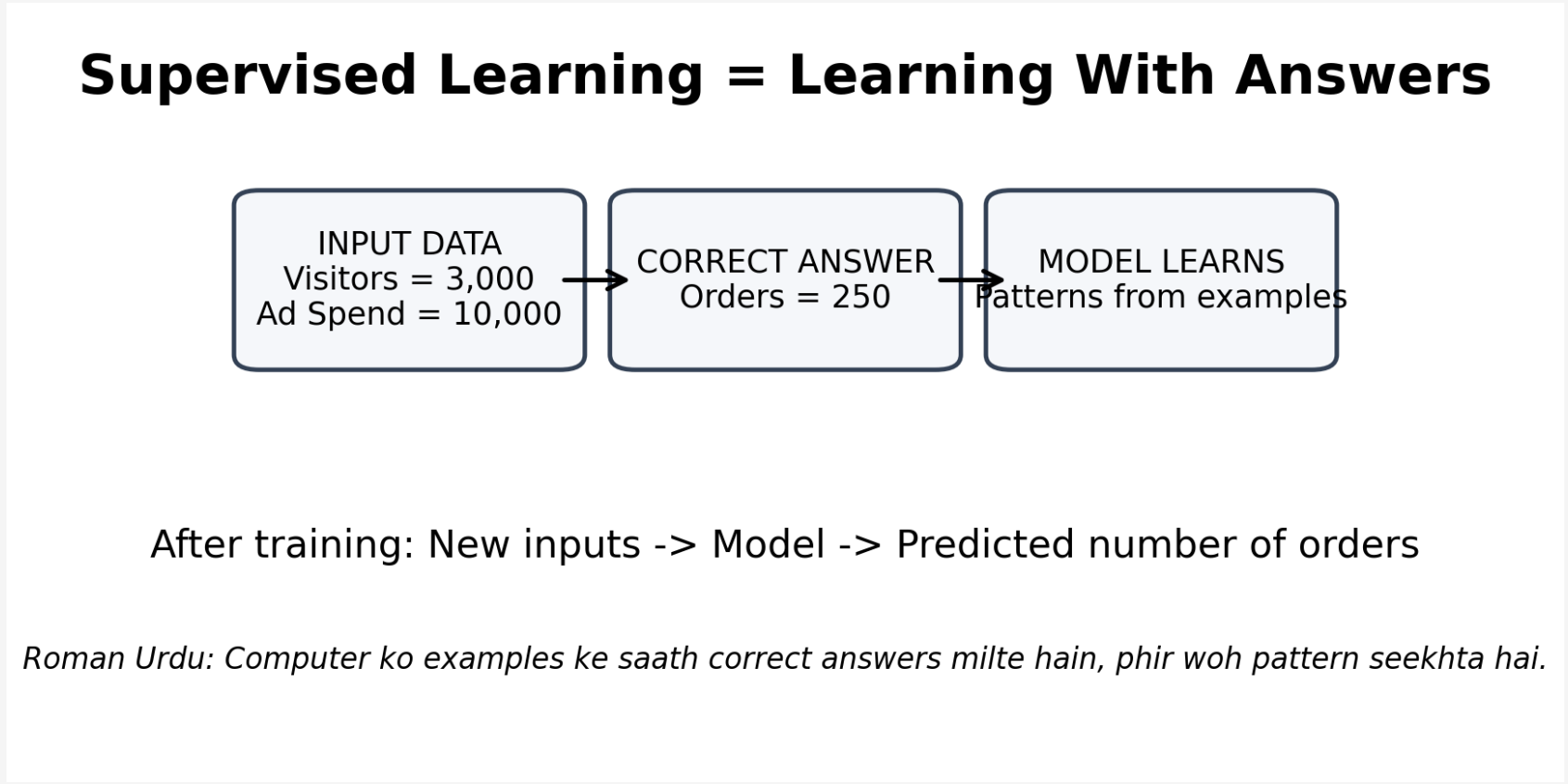
SEQUENCE MODELS

RNN → LSTM → GRU → Transformers

Today: Regression Goal: understand the complete real-world ML workflow before we move into neural networks.

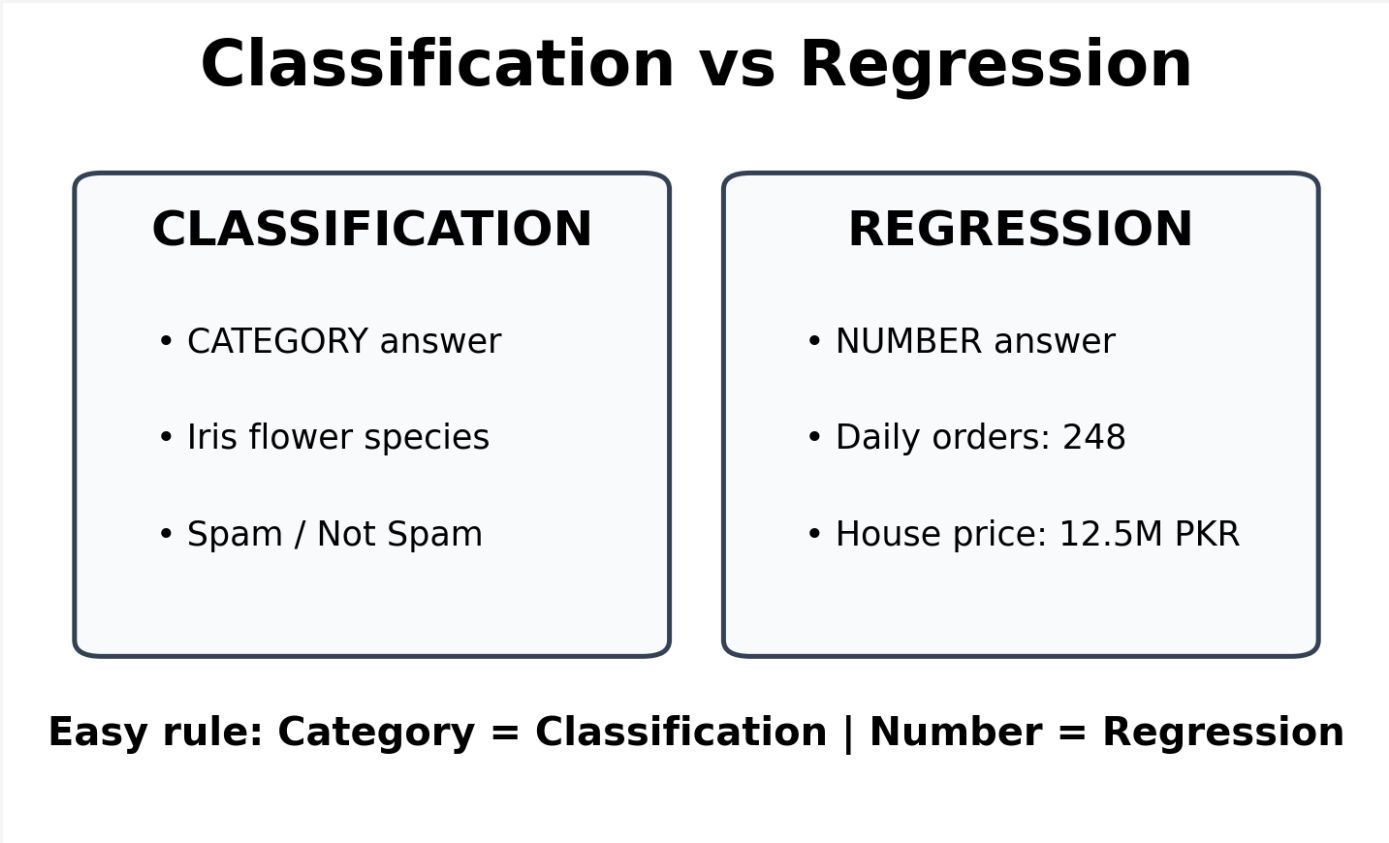
Supervised Learning = Learning With Answers

The model sees examples where the correct answer is already known during training.



Classification vs Regression

Both are supervised learning, but the type of answer is different.



Why Learn Regression?

Many real business questions ask “how much?” or “how many?”, not just which category.

Regression predicts a NUMBER

- Daily orders
- House price
- Sales revenue
- Delivery time
- Demand / quantity

Business value

- Plan stock
- Plan staff
- Set marketing budgets
- Forecast demand
- Compare scenarios

Our classroom question

“Based on today’s visitors, ad spend, discount and weekend flag…

how many orders should we expect?”

Project Overview: Daraz-Style Daily Orders Predictor

We will turn four business inputs into an expected number of daily orders.

INPUTS • Features

- Website Visitors
- Ad Spend (PKR)
- Discount (%)
- Weekend: 0 or 1

→

OUTPUT • Target

DAILY ORDERS

A continuous number
Example: 390 orders

This is a regression problem because the answer is a number.

Meet the Dataset

The project uses 120 synthetic daily records designed for learning regression.

Dataset Size

120

daily rows

6 columns

Synthetic classroom data

Columns

Day_Number	Day index / row sequence
Website_Visitors	Traffic to website
Ad_Spend_PKR	Marketing spend
Discount_Percent	Promotion level
Weekend	0 = weekday, 1 = weekend
Daily_Orders	TARGET: orders received

Rows, Features and Target

Machine learning starts by deciding what the model sees and what it must learn to predict.

ROW

One day

One historical business example

FEATURES (X)

Clues / Inputs

Visitors • Ad Spend
Discount • Weekend

TARGET (y)

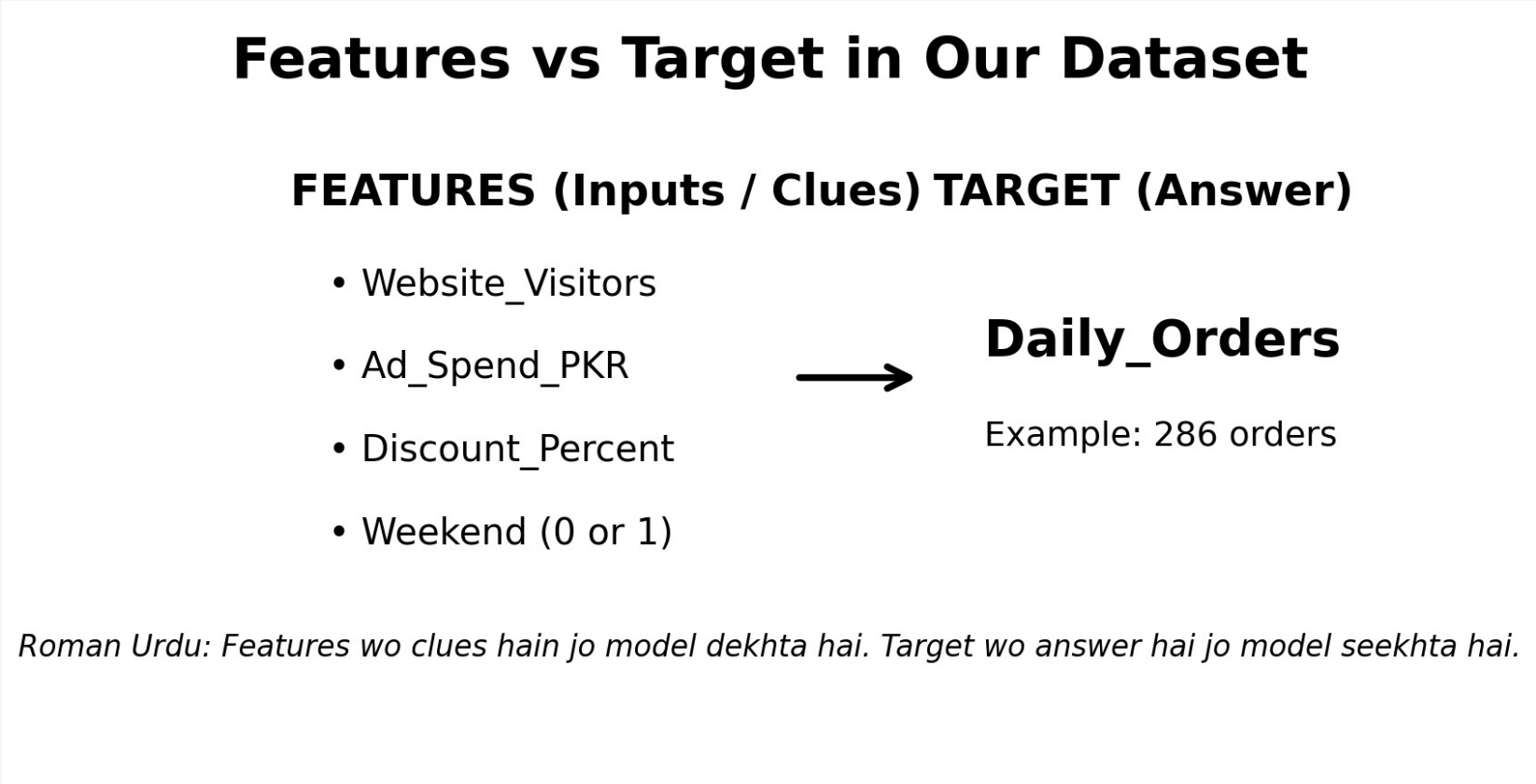
Daily Orders

The answer the model learns from

Supervised learning = features + known target during training.

Features vs Target in Our Dataset

Features are the clues. The target is the answer.



Why Are We NOT Using Day_Number as a Feature?

Choosing features is a modeling decision, not “use every column”.

In this classroom model

Day_Number is treated as an index of the synthetic records.

We intentionally use business drivers instead:

- Website Visitors
- Ad Spend
- Discount
- Weekend

Real-world note

A real date can contain useful information such as trend, season, holiday, month or day-of-week.

That would require meaningful feature engineering or a time-series approach.

Regression Intuition: Learn a Relationship

The model looks for a mathematical relationship between inputs and a numerical output.

LOWER INPUTS

2,000 visitors
5,000 PKR ads
5% discount
Weekday

≈ 170 orders

→

MEDIUM INPUTS

3,500 visitors
15,000 PKR ads
15% discount
Weekend

≈ 371 orders

→

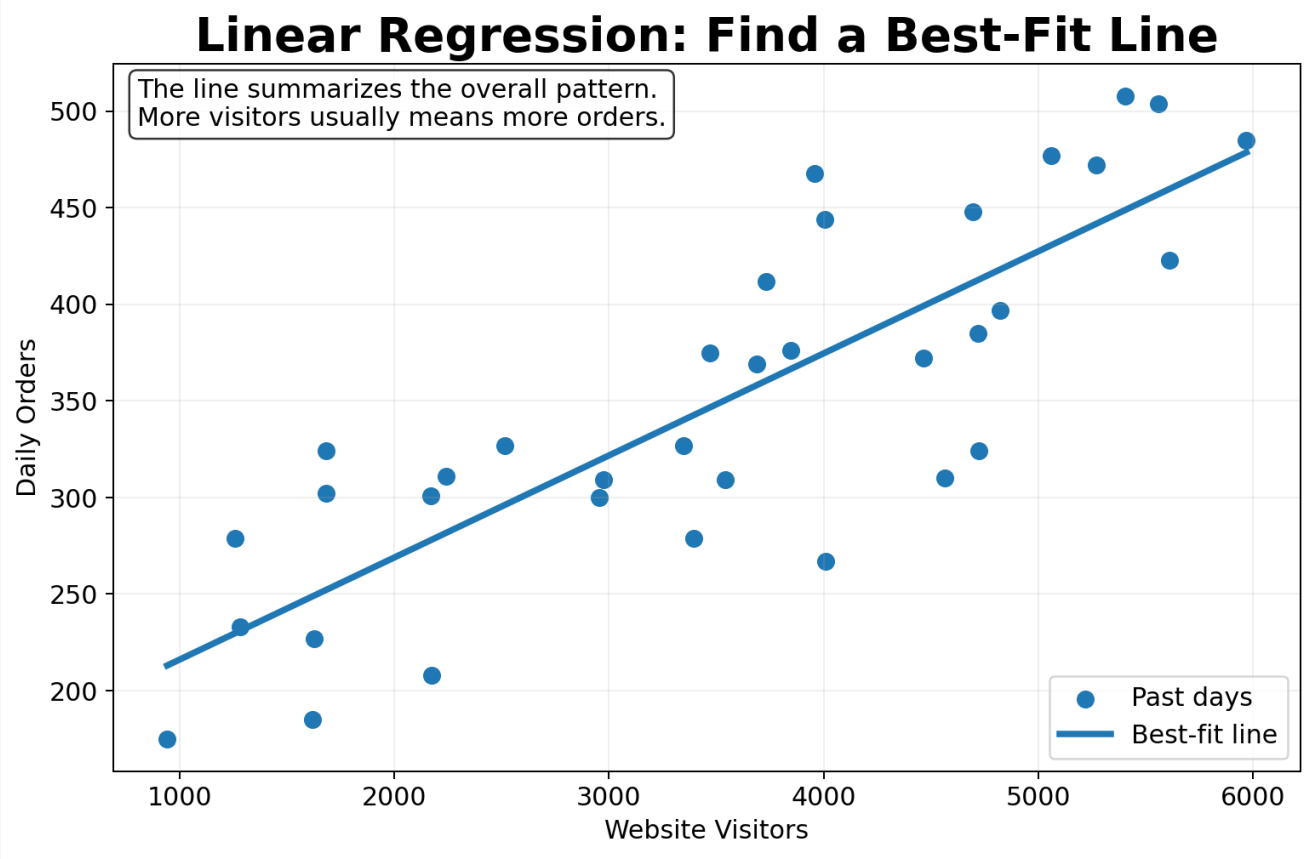
HIGHER INPUTS

6,000 visitors
25,000 PKR ads
25% discount
Weekend

≈ 580 orders

Linear Regression: Find the Best-Fit Line

The line summarizes the overall relationship instead of memorizing every individual point.



Simple vs Multiple Linear Regression

Our real project uses multiple inputs at the same time.

Simple Linear Regression

1 feature → 1 target

Example:
Website Visitors → Daily Orders

$$y = b_0 + b_1x$$

Multiple Linear Regression

4 features → 1 target

Visitors + Ad Spend + Discount + Weekend
→ Daily Orders

$$y = b_0 + b_1x_1 + b_2x_2 + b_3x_3 + b_4x_4$$

The Regression Equation: What Does Each Piece Mean?

Think of the equation as a recipe for combining feature effects.

Predicted Orders = Intercept + Visitor Effect + Ad Effect + Discount Effect + Weekend Effect

Intercept

starting baseline

Coefficient

effect per one unit

Feature value

today's input

Prediction

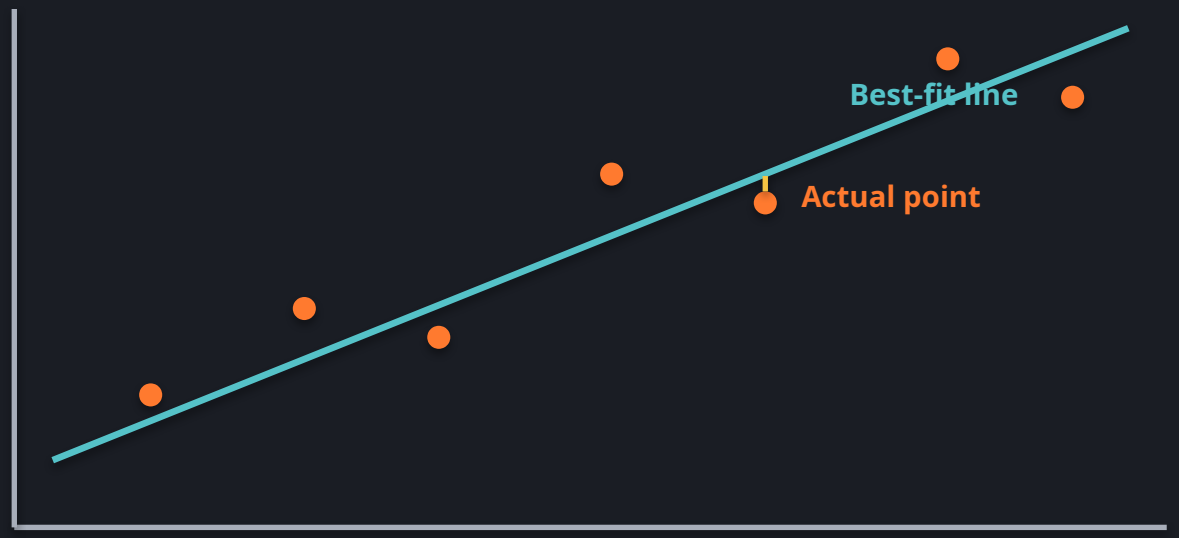
final estimated orders

The model learns the intercept and coefficients from training data.

How Does Linear Regression Learn? Minimize the Errors

Each point has an actual value. The line gives a predicted value. The vertical gap is the residual.

Residual idea



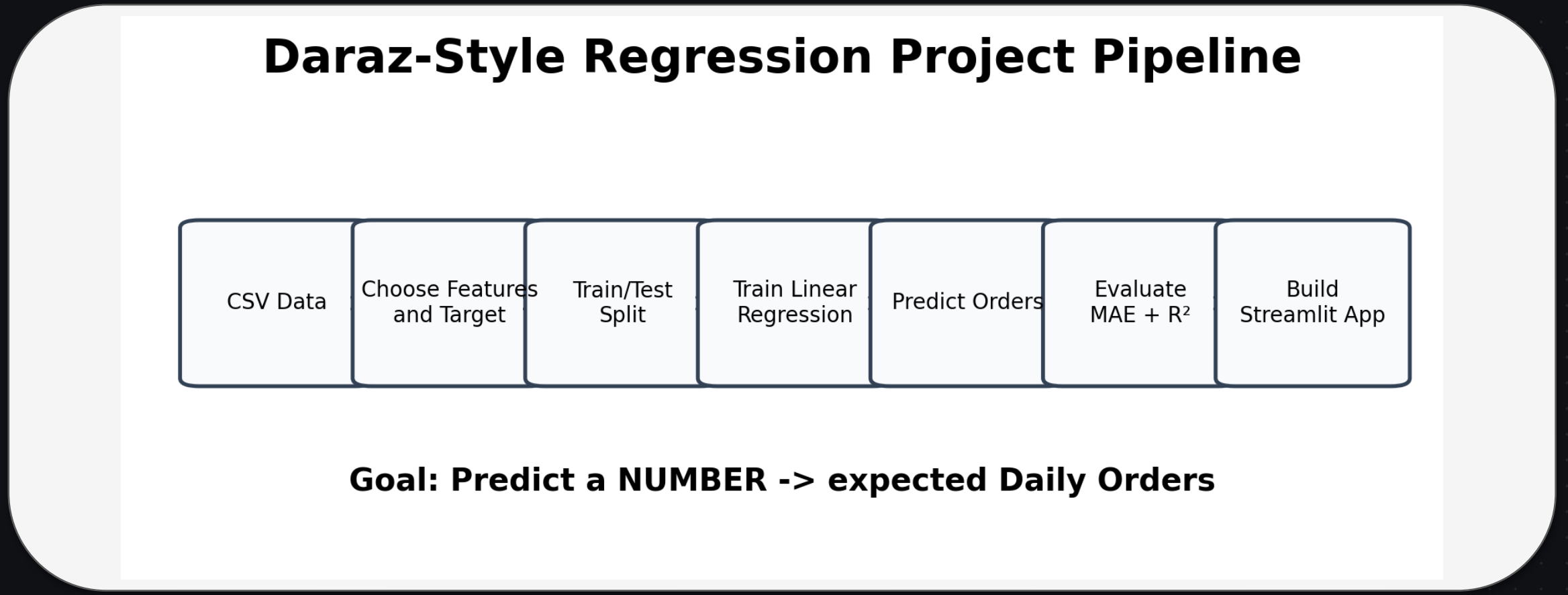
Training objective

- Make predictions
- Compare with actual orders
- Measure the errors
- Find coefficients that minimize overall squared error

Smaller overall residuals = better fit to training data

Our End-to-End Regression Pipeline

This is the complete project journey from CSV to a deployed prediction app.



Step 1: Load and Inspect the Data

Before modeling, understand the dataset instead of blindly calling model.fit().

CHECK SHAPE

120 rows × 6 columns

How much data do we have?

CHECK COLUMNS

Know every feature and the target.

Do not model a column you do not understand.

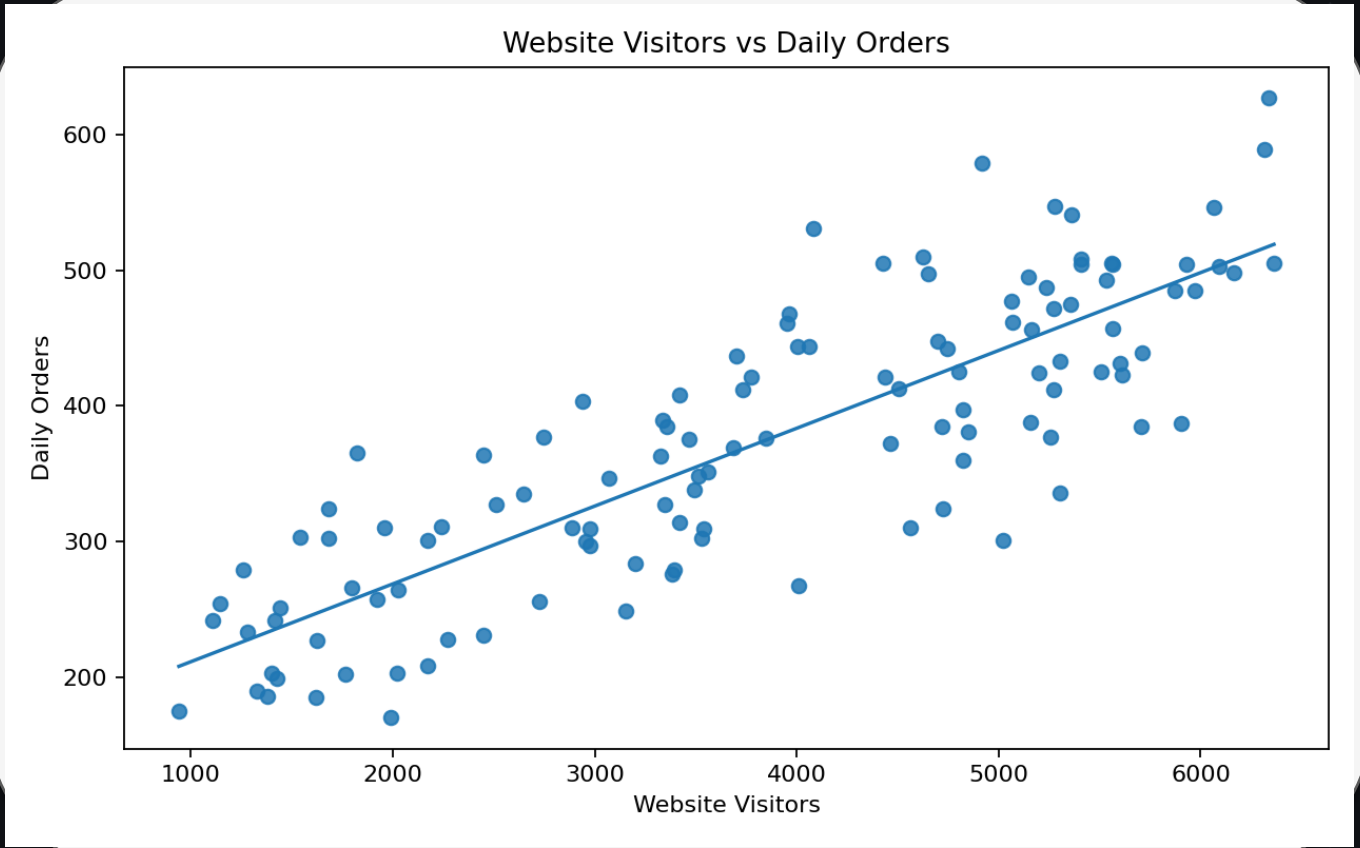
DESCRIBE NUMBERS

Minimum • Maximum
Mean • Spread

Use summary statistics to spot strange values.

Step 2: Explore Relationships Before Training

A scatter plot lets us visually ask: “Do more visitors usually mean more orders?”



What students should notice

- Points are not perfectly on one line
- Overall upward pattern exists
- Visitors are useful, but not the only driver
- That is why we add Ad Spend, Discount and Weekend

Step 3: Choose Features X and Target y

This is one of the most important decisions in supervised learning.

X = FEATURES

- Website_Visitors
- Ad_Spend_PKR
- Discount_Percent
- Weekend

The model is allowed to SEE these.

y = TARGET

Daily_Orders

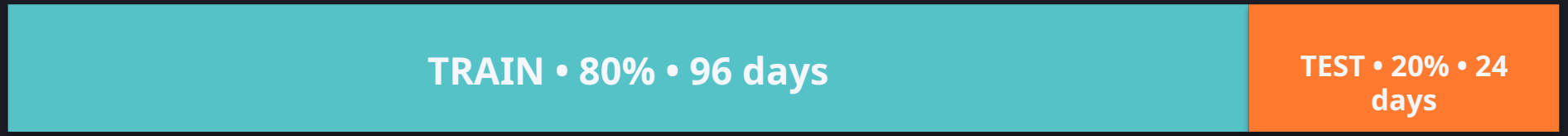
The model must LEARN to predict this.

Never accidentally put the target inside X. That would leak the answer.

Step 4: Train/Test Split

We keep part of the data hidden so we can test the model on unseen examples.

120 total days



Training set: learn coefficients

Test set: judge generalization

random_state = 42 makes the split reproducible.

Step 5: Train Linear Regression

Training means learning the intercept and coefficients that best fit the training data.



The algorithm chooses values that minimize squared prediction errors on training data.

No neural-network layers are involved here. This is classical machine learning.

What Did Our Model Learn?

These are the exact values learned from the 80/20 split in this project.

Intercept = 27.67

Website Visitors	**0.05091**	+100 visitors ≈ +5.09 predicted orders
Ad Spend (PKR)	**0.00392**	+1,000 PKR ≈ +3.92 predicted orders
Discount (%)	**4.181**	+1 percentage point ≈ +4.18 predicted orders
Weekend	**43.82**	Weekend = 1 ≈ +43.82 predicted orders

Interpretation is “holding the other model features constant” and describes the fitted association, not proven causation.

How to Read Coefficients Without Making a Wrong Claim

A coefficient is a model relationship, not automatic proof that one feature causes the outcome.

Safe interpretation

"Holding other included features constant, the fitted model associates a 1-point higher discount with about 4.18 more predicted orders."

Unsafe interpretation

"Every extra 1% discount CAUSES exactly 4.18 more orders."

Why unsafe? Real business data can contain hidden factors, interactions and confounding.

Step 6: Predict on the Test Set

The model has never trained on these 24 test rows. Now we compare prediction with reality.

ACTUAL

y_test

What really happened

→

MODEL

predict(X_test)

Estimated orders

→

COMPARE

Actual vs Predicted

Errors become evaluation metrics.

Prediction Error / Residual

For one row, error is the gap between actual orders and predicted orders.

Residual = Actual Orders − Predicted Orders

Actual 100

Predicted 90

Error 10

Actual 200

Predicted 220

Error -20 / absolute 20

Actual 300

Predicted 290

Error 10

Regression evaluation asks: how large are these errors, and how much variance does the model explain?

Metric 1: Mean Absolute Error (MAE)

MAE answers: on average, how many orders away are our predictions?

Simple example

Errors: 10, 20, 10

MAE = (10 + 20 + 10) ÷ 3

= 13.33

Our project

MAE = 23.55 orders

On the test set, predictions are about 23.55 orders away on average in absolute terms.

Lower MAE is better, but "good" depends on the business scale and baseline.

Metric 2: R^2 Score

R^2 tells us how much of the target variation is explained by the fitted model on the test data.

Interpretation

$R^2 = 1.0$	perfect fit on those points
$R^2 = 0.0$	no improvement over predicting the mean
$R^2 < 0$	worse than the mean baseline

Our project

$$R^2 = 0.945$$

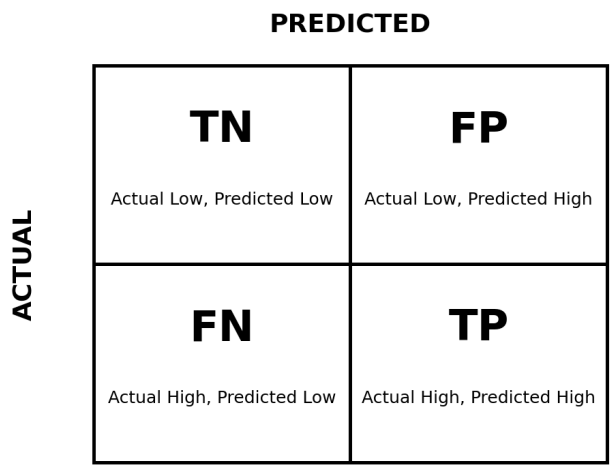
≈ 94.5% of the variation in test orders is explained by this fitted linear model.

Important: R^2 is NOT "94.5% accuracy".

Regression Metrics Are Different From Classification Metrics

We already met confusion matrix, precision and recall in classification. They do not judge continuous order predictions.

Confusion Matrix (Classification Only)



Important: We do NOT use a confusion matrix to judge a number-prediction regression model.

Precision and Recall

Example: Predicting HIGH SALES days

TP = 3 FP = 1 FN = 2 TN = 4

Precision = TP / (TP + FP) = 3 / 4 = 75%

Question: When we said HIGH, how often were we right?

Recall = TP / (TP + FN) = 3 / 5 = 60%

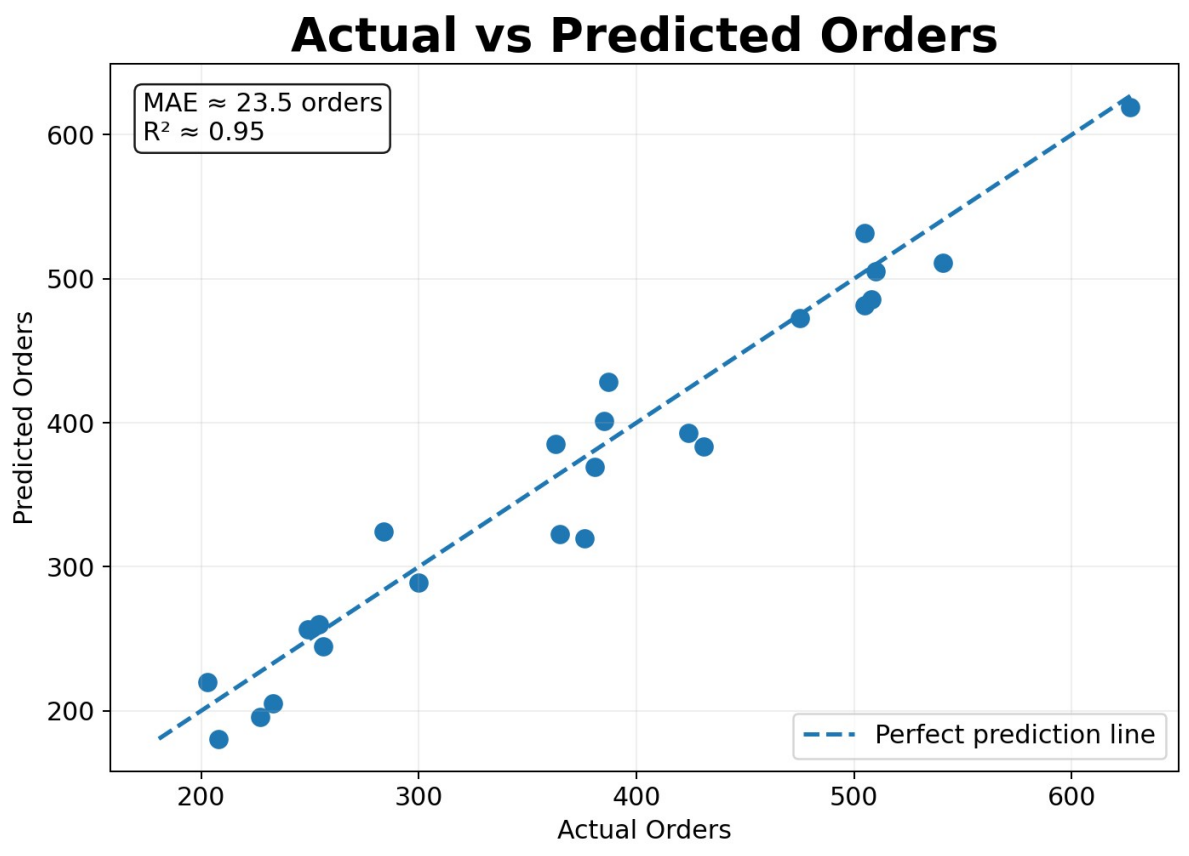
Question: Out of all truly HIGH days, how many did we find?

Classification: classes → confusion matrix / precision / recall

Regression: numbers → MAE / R^2 / RMSE etc.

Visual Evaluation: Actual vs Predicted Orders

Points close to the diagonal line mean predictions are close to actual values.



Predict New Business Scenarios

Once trained, the model can estimate expected orders for new combinations of inputs.

Scenario A

3,500 visitors

15,000 PKR ads

15% discount • Weekend

≈ 371 orders

Scenario B

2,000 visitors

5,000 PKR ads

5% discount • Weekday

≈ 170 orders

Scenario C

6,000 visitors

25,000 PKR ads

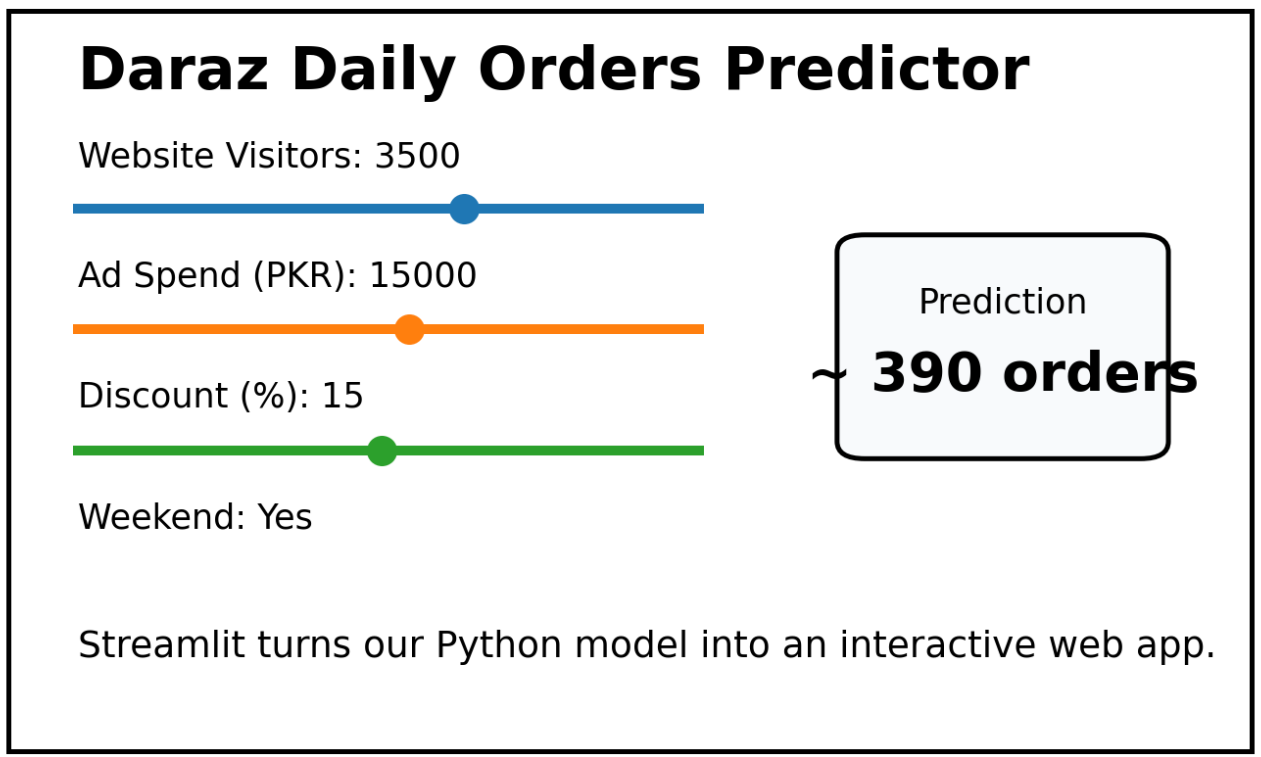
25% discount • Weekend

≈ 580 orders

These are model estimates, not guaranteed future sales.

Turn the Model Into a Streamlit Web App

A user should not need to open a notebook just to make one prediction.



Portal flow

- User enters business inputs
- App builds one-row DataFrame
- Model predicts orders
- Result displayed instantly
- No retraining for every click

Streamlit = Python model → interactive web interface

From Local Project to GitHub and Deployment

A complete ML project includes reproducible files, not only a trained model in memory.

1

Project files

app.py • CSV • requirements

2

Local test

streamlit run app.py

3

GitHub

push repository

4

Deploy

Streamlit Community Cloud / chosen hosting

Deployment means other people can use the model through a browser.

What Can This Regression Skill Be Used For?

The dataset changes, but the supervised-regression thinking transfers to many industries.

E-COMMERCE

Daily orders
Revenue
Demand

BANKING

Deposit amount
Transaction volume
Customer value

LOGISTICS

Delivery time
Shipment volume
Fuel usage

REAL ESTATE

House price
Rent estimate
Property value

OPERATIONS

Call volume
Workload
Resource demand

MARKETING

Sales uplift
Leads
Campaign response volume

A Good Data Scientist Also Knows the Limitations

A high R^2 on a classroom dataset does not automatically mean a production-ready forecasting system.

Current project limits

- Synthetic data
- Only 120 rows
- Linear relationships only
- No holidays / seasonality
- No inventory or competitor effects

What could improve it?

- More real historical data
- Date / season features
- More business drivers
- Cross-validation
- Compare multiple models

Critical thinking

Never confuse correlation with causation.

Always compare with a baseline and validate on unseen data.

Production models also need monitoring and governance.

Machine Learning vs Deep Learning: What Changes Next?

This regression project prepares students for ANN, but classical ML and deep learning are not the same thing.

Classical Machine Learning

- Example: Linear Regression
- Direct mathematical model
- Few parameters
- Fast on small tabular data
- Highly interpretable coefficients

Deep Learning

- Example: Artificial Neural Network
- Layers of connected neurons
- Many trainable weights and biases
- Activation functions + backpropagation
- Can learn more complex nonlinear patterns

Next module: we keep supervised learning, but replace the linear model with a neural network.

What You Must Understand Before ANN

Regression is not just a model. It is an end-to-end way of thinking about a numerical prediction problem.

- Supervised learning = examples with known answers
- Classification predicts categories; regression predicts numbers
- Features X are clues; target y is the answer
- Train/Test split protects honest evaluation
- Linear Regression learns intercept + coefficients
- MAE measures average absolute error
- R^2 measures explained variation, not accuracy
- A Streamlit app turns the model into a usable tool

NEXT: Artificial Neural Networks → Deep Learning